

CCNA
Cisco Certified Network Associate

Exam code

200 - 301

Switch

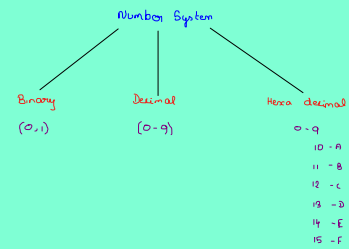
Router

AP

Firewall

Local → Small network

Global → Cisco 30K, 35K



Base value :-

Total number in a number system is called
Base value.

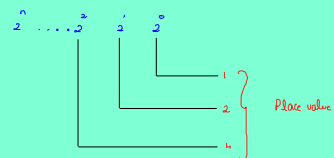
Binary = 2

Decimal = 10

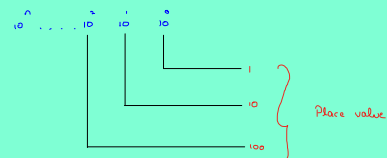
Hexa = 16

Place value :

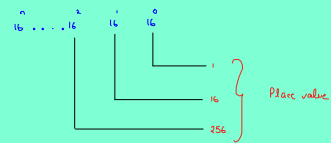
Binary :



Decimal :



Here



Conversions

1 Octet = 8 bits

128 64 32 16 8 4 2 1

Decimal - Binary

128 64 32 16 8 4 2 1

10 0 0 0 0 1 0 0
20 0 0 0 1 0 1 0
192 1 1 0 0 0 0 0
224 1 1 1 0 0 0 0
255 1 1 1 1 1 1 1

1 - count
0 - don't count

Binary - Decimal

128 64 32 16 8 4 2 1

1 1 0 1 0 1 1 235
1 1 0 0 1 0 1 203
1 1 0 1 0 1 0 212
1 1 0 1 1 0 1 211
1 0 1 0 1 1 0 133

Hexa - Binary

8 4 2 1

A 1 0 1 0
B 1 0 1 1
C 1 1 0 0
D 1 1 0 1
E 1 1 1 0
F 1 1 1 1

1 Hexa = 4 bits

8 4 2 1

8 4 2 1

0 0 0 0
0 0 0 1
0 0 1 0
0 0 1 1
0 1 0 0
0 1 0 1
0 1 1 0
0 1 1 1
1 0 0 0
1 0 0 1
1 0 1 0

Physical connection

Draw back

- * Time
- * Cost
- * Security

Solution

Network

What is network ?

Two or more devices (computers, mobiles, servers, printers, etc.) interconnected with each other for sharing resources, information, and communication

medium

Radio Frequency } wire less

Electrical Signal } wired

Light Signal } fiber optic cable

UTP Unshielded twisted Pair

It contain 4 pair

Totally it contain 8 copper wire

Crimping

1 - G

2 - B

3 - G

4 - B

5 - W

6 - B

7 - G

8 - B

Crimping method

- * Straight cable
- * Crossover cable

T568A

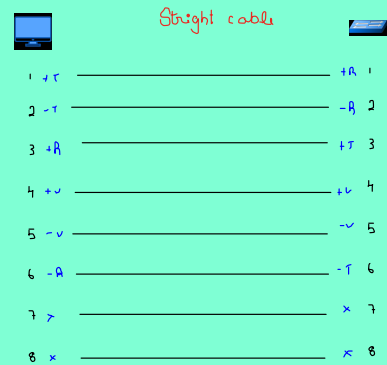
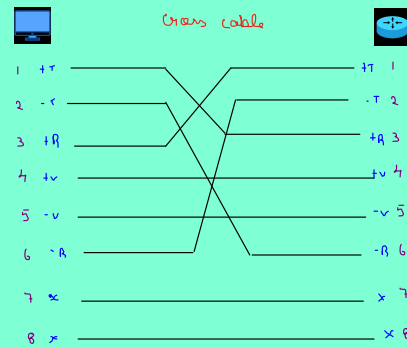
T568B

T568B

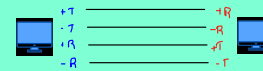
T568A

Title: Jul 12-9:36 AM (3 of 4)

DEVICE	+T -T	+R -R
PC	1 2	3 6
ROUTER	1 2	3 6
SWITCH	3 6	1 2



Auto mix



NIC

- Network interface card
- It contains mac + ip

Function of NIC

It is use to convert binary signals to electrical & electrical signals to binary

Types of nic

- * wired
- * wireless